

CampusConnect

Feasibility Report & Software Requirements Specification (SRS)

Prepared By : Mhirit Hiba, Mana Salma

Standard Specification: IEEE 830-1998

Platform Target : mobile app

Document Status : Final Feasibility & Requirements Submission

Executive Summary

Project Objective: *CampusConnect connects students with the information, resources, things, skills, and people they need within their university community.*

Table of Contents

Section

Part I: Feasibility Report

1. Technical Feasibility
 2. Schedule Feasibility
- Conclusion

Part II: Software Requirements Specification (SRS)

1. Introduction
2. Overall Description
3. Specific Requirements
 - 3.1 Functional Requirements
 - 3.2 Non-Functional Requirements
4. Appendices & 5. Index

Part I: Feasibility Report

1. Technical Feasibility

CampusConnect is technically feasible as a mobile app. The team already knows React.js, HTML/CSS, MySQL, Git/GitHub and Draw.io. The main technologies to learn are Java Spring Boot, REST APIs and JPA/Hibernate.

Technology / Tool	Status / Use
React.js, HTML/CSS	Already known – frontend
MySQL	Already known – database
Git/GitHub, VS Code, Draw.io	Already known
Java + Spring Boot	Java is being learned; Spring Boot is new – backend
JPA/Hibernate + REST API	New – database and frontend/backend communication
Postman	To learn – API testing
Laptop + browser + Internet	Required

Main challenges: learning Spring Boot, connecting React to the Java backend, connecting Spring Boot to MySQL, and implementing basic authentication.

2. Schedule Feasibility

The project is planned over approximately 8 weeks, with 1–2 working days available per week.

Week	Main task
1	Requirements, UML, database design, UI planning
2	Spring Boot setup and REST API basics
3	MySQL + JPA/Hibernate + User/Post entities
4	Authentication + NEED/HAVE posts
5	Search, filters + matching
6	Requests + basic messaging
7	Ratings + UI improvements
8	Testing, bug fixing, documentation + presentation

The main dependency is learning Spring Boot and JPA early. The final week should be kept for testing, fixing bugs and preparing the final submission.

Conclusion

CampusConnect is feasible within the available time because most of the required technologies are already known.

Part II: Software Requirements Specification

1. Introduction

Purpose: CampusConnect is a platform that helps students find campus information, resources, objects, skills, and student-to-student assistance in one place.

Scope: The first version targets one university. Students can:

Find campus information

Post I NEED requests

Post I HAVE offers

Exchange skills

Find compatible students

Communicate and schedule exchange.

Review and rate other students.

2. Overall Description

2.1 Product Perspective: CampusConnect is a mobile application connected to a backend and database.



The application centralizes student needs and offers within the campus community.

External Interfaces:

User: Mobile interface

Hardware: Smartphone

Software: Mobile OS, backend, database.

Communication: Internet using HTTP/HTTPS.

2.2 Product Functions: The main functions are:

User Management: registration, login, profile.

Campus Information: procedures, facilities, transport, study places, resources.

I NEED: students request something they need.

I HAVE: students offer something they have.

Skills Exchange: teach and learn skills.

Matching: connect compatible students.

Messaging: communicate with matching students.

Scheduling: arrange meetings/exchanges.

Reviews: rate interactions.

Administration: manage users and campus content.

2.3 User Characteristics:

Student: Main user of the application. Basic smartphone skills. No advanced technical knowledge required.

Administrator: Manages users and campus information. Moderates inappropriate content. Requires basic computer and administrative skills.

2.4 Constraints:

- Initially designed for one university.
- Requires a compatible smartphone.
- Online features require Internet access.
- Must use secure communication such as HTTPS.
- Only authorized users can access protected information.
- Database, backend technology, mobile platform, and development standards constrain implementation choices.

2.5 Assumptions and Dependencies:

Assumptions

- Students have smartphones and Internet access.
- Users provide accurate information.
- University information is accurate and maintained.

Dependencies

- Backend server.
- Database.
- Internet connection.
- Mobile operating system.

3. Specific Requirements

3.1 Functional Requirements:

FR1	The system shall allow students to register and log in.
FR2	The system shall allow students to view campus information.
FR3	The system shall allow students to create an I NEED request.
FR4	The system shall allow students to create an I HAVE offer.
FR5	The system shall allow students to search requests and offers.
FR6	The system shall allow students to publish and search skills.
FR7	The system shall identify compatible matches.
FR8	The system shall allow matched students to communicate.
FR9	The system shall allow students to schedule exchanges.
FR10	The system shall allow students to rate and review other students.
FR11	The system shall allow administrators to manage users and campus information.

Use Case Diagram Details:

Main Actors:

- Student.
- Administrator.

Student Use Cases:

- Register / Login.
- Manage Profile.
- View Campus Information.
- Create I NEED.
- Create I HAVE.
- Search.
- Exchange Skills.
- Find Matches.
- Message.
- Schedule.
- Rate / Review.

Administrator Use Cases:

- Manage Users.
- Manage Campus Information.
- Moderate Content.

Data Flow Diagram Structural Overview:

Student / Admin —► CampusConnect —► Database

The system exchanges information concerning users, posts, skills, matches, messages, sessions, and reviews.

3.2 Non-Functional Requirements:

NFR1: The system should support the expected number of university users.

NFR2: Protected information shall require authentication.

NFR3: Communication shall use HTTPS.

NFR4: Administrative functions shall be restricted to administrators.

NFR5: Passwords shall be securely hashed.

NFR6: The interface shall be simple and consistent.

NFR7: The system shall provide meaningful error messages.

4. Appendices & 5. Index

Appendices May Include: Complete Use Case Diagram, Data Flow Diagram, Database/ER Diagram, UI mockups, Requirements traceability table, Test cases.

Index Important Terms: Administrator, Campus Information, I HAVE, I NEED, Matching, Student, Functional Requirements, Non-Functional Requirements.